Category: Cybersecurity

PS-024: Privacy Pass Token Issuance and Anti-Replay Gateway

- **Description:**
  - Traditional web rate limiting relies on client IP tracking, which fails when users browse through shared proxies, VPNs, or Apple iCloud Private Relay, where multiple clients share one egress address. The IETF Privacy Pass architecture (RFCs 9576 to 9578) uses cryptographic blind signatures so clients can redeem anonymous authorization tokens without linking redemption to issuance. In this project, you will integrate an existing Privacy Pass library into a lab issuer, client, and edge verification gateway. The gateway will redeem tokens for an origin and maintain an anti-replay cache to reject reused or invalid tokens, measuring verification latency and operational overhead across a shared-proxy test.
- **Challenge:**
  - Integrate an existing Privacy Pass library into a lab issuer, client and verification gateway. Demonstrate token issuance, verification and single redemption while evaluating unlinkability assumptions and operational cost.
- **Key Requirements**
  - Implement 1 supported RFC 9578 issuance protocol.
  - Use 1 controlled origin with isolated issuer and origin keys.
  - Implement issuer, client and verification gateway components.
  - Enforce single redemption using a replay cache.
  - Reject reused, invalid and wrong-context tokens.
  - Test the system through a shared-proxy scenario.
  - Measure token issuance and redemption latency.
  - Identify metadata retained by the issuer, client and gateway.
  - Treat ARC as an optional separate extension.
  - No new cryptography, guaranteed sliding-window anonymous quotas, or claim that Privacy Pass hides transport-layer IP addresses.
- **Expected Outcome**
  - End-to-end Privacy Pass token demo
  - Issuer and verification gateway integration
  - Client implementation
  - Replay-cache mechanism
  - Invalid, reused and wrong-context token tests
  - Shared-proxy test scenario
  - Threat model and unlinkability analysis
  - Issuance/redemption latency measurements
  - Metadata-retention analysis
- **Deliverables***
  1. GitHub Collaboration: (**This step is crucial for allowing mentors to provide real-time oversight.**)
     - Each team must set up a private repository on GitHub.

- Add our GitHub account (https://github.com/aiori-hackathon) as a collaborator.
- Once the collaboration request has been accepted all the concerned users (all the teammates as well as aiori-hackathon account) transfer the ownership of the repository to our GitHub account. (https://github.com/aiori-hackathon)
- Ensure that your repository is named according to your team's name.

2. Each team is required to upload a PDF to their GitHub repository
   - Please follow the specified structure: Proposed-structure-hackathon.pdf
3. Prototype Demonstration with a Pseudocode Snippet to the mentor

- **References**
  - https://datatracker.ietf.org/doc/rfc9576/